

DOSM Datathon 2026

**GEOPOLITICAL RISK AND INTERNATIONAL TOURISM DEMAND IN
MALAYSIA: A MACHINE LEARNING FORECASTING FRAMEWORK FOR 20
SOURCE MARKETS**

by

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1.0 INTRODUCTION

1.1 Background of the Study

Tourism is a major part of Malaysia's economy and labour market. Tourism industries generated RM323.0 billion in value added in 2025, equivalent to 15.9% of gross domestic product, and tourism-related industries employed 3.7 million people (Department of Statistics Malaysia [DOSM], 2026b). International demand is central to this contribution. Malaysia received 25,016,698 tourists in 2024, 24.2% more than in 2023 but still 4.2% below 2019 (Tourism Malaysia, 2025). Singapore, Indonesia and China were among the largest source markets. Malaysia's tourism performance therefore depends partly on economic and political conditions affecting travellers and the routes that connect them to the country.

Geopolitical risk covers the threat, occurrence and escalation of wars, terrorism and tensions among states or political actors. Caldara and Iacoviello (2022) measure it through a news-based Geopolitical Risk Index. For tourism, higher risk may alter perceptions of safety, prompt travel advisories or destination substitution, disrupt air routes and raise transport costs through energy-price movements. Exchange-rate changes can also alter Malaysia's affordability for travellers from different countries. These channels suggest that the relationship may vary by source market and over time.

Research generally links higher geopolitical risk with weaker tourism demand, but results differ among countries. Lee et al. (2021) found a negative long-run relationship in a 16-country panel, yet their country-level tests did not identify a significant causal relationship between geopolitical risk and Malaysia's inbound tourist numbers. Papagianni et al. (2024) also found heterogeneous responses across emerging and developing economies. An average international result should therefore not be transferred directly to Malaysia without a current, source-market-specific analysis.

This project examines monthly tourist arrivals from 20 modelled source markets. Its analytical window runs from March 2017 to September 2026, with published arrivals through May 2026 and June to September treated as the forecasting horizon. The country-month panel combines arrivals with bilateral exchange rates, global and Malaysia-specific geopolitical risk, Brent crude oil and Malaysian fuel prices. Lagged arrivals and seasonal features represent established demand patterns. The resulting structure tests whether geopolitical information contributes to short-term forecasts after these factors are considered.

1.2 Problem Statement

Existing evidence on the geopolitical risk–tourism relationship is inconsistent across countries, and the study most directly relevant to Malaysia found no significant relationship at the country level (Lee et al., 2021), even though the same relationship is negative on average internationally. This inconsistency means an average international finding cannot be assumed to hold for Malaysia, and no published study appears to test the relationship separately for each of Malaysia's individual source markets rather than for Malaysia as a single national aggregate. This aggregation problem is compounded by a forecasting gap. Malaysian tourism research to date has mainly examined aggregate or long-run relationships between geopolitical risk and tourist arrivals, while short-term, source-market-level forecasts that integrate monthly arrivals with exchange rates, geopolitical risk and energy prices – built under a validation design that avoids data leakage and respects real publication-timing constraints – are not available in the literature (see Section 2.5). Existing Malaysian tourism reporting typically presents a single national arrivals figure, which can mask divergent conditions across the 20 major source markets that supply the bulk of international demand.

Without market-level information, decision-makers responsible for marketing, capacity planning and workforce allocation cannot distinguish a broad-based decline in demand from one concentrated in a small number of origin countries, nor can they anticipate near-term changes with enough lead time to act. A forecasting and monitoring tool that operates at the source-market level, discloses its own publication-timing and data limitations, and separates statistical association from causal claims is therefore needed to support more targeted and defensible tourism-policy decisions.

This study addresses this problem by evaluating, market by market, whether geopolitical risk carries forecasting information for Malaysia's 20 largest international source markets once lagged arrivals, seasonality, exchange rates and energy costs are already accounted for, and by translating that evaluation into a short-term, one-to-four-month forecasting and monitoring framework whose accuracy, limitations and assumptions are reported transparently.

1.3 Research Objectives

The general objective of this study is to evaluate the predictive relationship between geopolitical risk and international tourism demand in Malaysia and to develop an interpretable short-term forecasting framework for the 20 modelled source markets.

1. To construct a traceable monthly country panel that integrates tourist arrivals, bilateral exchange rates, global and Malaysia-specific geopolitical risk, Brent crude oil prices and Malaysian fuel-price indicators for the period from March 2017 to September 2026.
2. To assess how geopolitical-risk variables contribute to source-market arrival patterns after controlling for lagged demand, seasonality, exchange rates and energy costs, while interpreting all results as non-causal associations.
3. To develop one-to-four-month arrival forecasts using a pooled direct multi-horizon LightGBM model and compare its performance with naive, seasonal-naive and Ridge-regression benchmarks through rolling-origin MAE and RMSE evaluation.
4. To test the robustness of the forecasting framework under alternative arrival-publication-lag assumptions and a separate COVID-period stress test, then communicate the results through a decision-support dashboard for tourism monitoring and planning.

1.4 Scope and Significance of the Study

The study covers 20 selected source markets and uses the country-month as its analytical unit. Aggregated results must not be labelled as Malaysia's total international arrivals because other markets are excluded. Published arrivals end in May 2026; June to September form the live forecast period. The repository's annual 16-state hospitality panel supplies wider context but is not an input to the final international-arrivals model.

The project links Malaysian tourism and economic data with external indicators that may transmit geopolitical conditions. Source-market forecasts can reveal changes hidden by national totals, while the dashboard supports monitoring, market diversification and targeted planning. Rolling-origin evaluation estimates how the model would have performed using information available at each historical forecast date.

The conclusions remain bounded. News-based GPR indices may move with other global events, especially during the pandemic and recovery. The model estimates associations, not causal effects, and each forecast uses predictors available at the forecast date. The latest forecasts should therefore be interpreted as early-warning estimates rather than precise predictions or proof that geopolitical risk alone caused changes in Malaysian tourism demand.

2.0 LITERATURE REVIEW

2.1 Geopolitical risk and tourism demand

Geopolitical risk covers threats and realised events involving war, terrorism and interstate tension. Caldara and Iacoviello's (2022) news-based index distinguishes these dimensions. For tourism, such risk can alter perceived safety, travel advisories, air routes, operating costs and destination choice, including when conflict occurs outside Malaysia. Global conflicts may also affect household confidence and energy prices before changing observed travel demand.

Evidence is generally negative but heterogeneous. Lee et al. (2021) found lower long-run tourism demand across 16 countries, yet no causal relationship between GPR and Malaysian tourist numbers; Malaysian tourism receipts showed a different pattern. Papagianni et al. (2024) also reported persistent negative responses in most of 14 emerging economies, with stronger historical effects in China, Indonesia and Thailand. The size, timing and direction of the response therefore cannot be assumed uniform across destinations or source markets.

Malaysia combines neighbouring land markets with long-haul markets dependent on air connectivity. National totals may conceal differences in income, trip purpose and exposure to particular events. This study consequently models monthly arrivals from 20 source markets and includes global GPR, Malaysia-specific GPR and country interactions to allow the controlled association to vary across markets. This design can distinguish a broad international shock from a response concentrated in selected origins.

Because GPR also moves with economic conditions, energy prices, post-pandemic recovery and omitted factors, prediction does not establish causation. The model uses GPR as a forecasting signal, while Ridge coefficients and SHAP values are interpreted as model-based associations only.

2.2 Oil Prices, Transport Costs and Tourism Demand

Transport is part of a trip's total price, and oil affects both aviation and road costs. Higher transport expenditure can also reduce the budget available at the destination. Becken and Lennox (2012) linked sustained oil-price increases to lower tourism exports through transport, income and exchange-rate channels. Across 23 European countries, Boto-Garcia (2025) estimated an average arrival elasticity near -0.2 and persistent effects from supply-driven oil shocks. Although context-specific, this supports energy prices as short-term predictors.

Oil nevertheless imperfectly represents traveller costs because airlines hedge fuel, vary surcharges and adjust capacity, while land travellers may face regulated retail prices. Seetaram

(2010) warns that oil, distance and published airfares introduce measurement error. Tan and Soon (2023) likewise found that Malaysian travel-cost responses differed by region rather than showing a uniform demand reduction.

In 2024, 41.0% of Malaysian tourists arrived by air and 52.4% by land (Tourism Malaysia, 2025). The model therefore combines Brent crude as global transport-cost pressure with Malaysian retail fuel prices as a road-travel indicator. This captures two possible transmission channels without assuming that crude-price changes reach every market equally or immediately. The distinction is especially relevant to Malaysia's mix of regional land arrivals and long-haul aviation demand.

2.3 Exchange Rates and Malaysia's Destination Affordability

Exchange rates determine how much visitors can buy in Malaysia with origin-country income. Ringgit depreciation should improve affordability, whereas appreciation reduces price competitiveness. Volatility can also create uncertainty about the final trip cost, delaying bookings, shortening stays or encouraging destination substitution. Tourism-demand reviews identify income, destination prices and bilateral exchange rates as core variables (Song & Li, 2008), while disaggregated evidence shows that level and volatility effects differ across origins (Imamboccus et al., 2024).

Malaysian evidence rejects a simple weaker-ringgit-equals-more-tourists rule. Karimi et al. (2019) found asymmetric effects, with both appreciations and depreciations reducing arrivals in some specifications, which they linked to price rigidity and substitution. Across 21 origins, Tan and Soon (2023) found depreciation increased ASEAN demand but reduced demand from China, other Asian markets and Western countries. Income, airfares, competing destinations and incomplete price pass-through may further shape these responses.

Bilateral rates are therefore preferable to an aggregate index: one ringgit movement changes affordability differently for visitors paid in Singapore dollars, yuan or US dollars. The pooled model preserves this variation while sharing information across markets. Exchange rates are treated as market-specific affordability indicators rather than isolated causal determinants. This avoids masking opposing origin-market responses.

2.4 Malaysian Tourism Resilience, Economic Conditions and SDG Alignment

Malaysia recorded 25,016,698 tourist arrivals in 2024, 24.2% above 2023 but 4.2% below 2019 (Tourism Malaysia, 2025). In 2025, tourism generated RM323.0 billion in value added, or 15.9% of GDP, and supported 3.7 million jobs (Department of Statistics Malaysia [DOSM],

2026b). These figures show both the recovery and the sector's economic scale. Forecast errors therefore affect staffing, transport, marketing and the timing of public support.

Recovery has not removed vulnerability. COVID-19 exposed labour precarity and complex supply-chain weaknesses in Penang (Hampton et al., 2023), while dependence on a few source markets increases exposure to health, economic and geopolitical shocks. Country-level forecasts reveal whether weakness is broad or concentrated, enabling earlier adjustments to marketing, connectivity and workforce plans.

This decision-support role supports the National Tourism Policy 2020-2030 emphasis on competitiveness, sustainability, inclusiveness, smart tourism and responsible tourism (Ministry of Tourism, Arts and Culture Malaysia [MOTAC], 2020). It also relates to SDG target 8.9, covering sustainable tourism, jobs and local culture, and target 12.b, which calls for tools that monitor tourism's sustainability impacts (United Nations Department of Economic and Social Affairs [UN DESA], n.d.).

Forecasting itself does not prove environmental or social sustainability. Its contribution is operational: timely market-level information can reduce avoidable over- or under-capacity and support planning for volatility. The dashboard should show forecast coverage and limitations, while decisions combine predictions with environmental, community and labour indicators rather than equating higher arrivals with sustainable progress. This is important because tourism performance affects both national output and a large tourism-related workforce.

2.5 Machine-Learning Approaches and Research Gap

Tourism demand is seasonal, persistent and shock-sensitive. Traditional methods include naive and seasonal-naive forecasts, exponential smoothing, autoregressive models and econometric equations. Machine learning can capture nonlinear interactions among lags, seasonality, prices and external indicators, but no method dominates every destination or horizon (Song & Li, 2008). Forecast accuracy also varies with the origin, destination, sample, frequency, variables and horizon (Peng et al., 2014), so models require task-specific out-of-sample testing.

Pooling lets short market series share information through one global forecasting function. Global models can generalise across related series and are not inherently more restrictive than separate local models (Montero-Manso & Hyndman, 2021). However, tourism accuracy suffers if market heterogeneity is ignored; fixed, spatial or temporal effects can improve performance (Long et al., 2019). This study therefore adds country identifiers, market features and GPR-by-country interactions rather than assuming identical responses.

LightGBM captures nonlinear thresholds and mixed-feature interactions (Ke et al., 2017) and performed effectively with correlated series and exogenous variables in the M5 competition (Makridakis et al., 2022). However, retail success does not guarantee tourism superiority. Ridge regression provides a transparent regularised comparison, while naive-last and seasonal-naive forecasts remain essential benchmarks. This comparison tests whether added complexity improves on simple historical rules. SHAP assigns feature contributions to predictions but explains model behaviour rather than causal effects (Lundberg & Lee, 2017).

Rolling-origin evaluation reproduces repeated real-time decisions and measures each lead time without random-split leakage (Tashman, 2000). Separate horizon results are important because accuracy may decline as the forecast extends. MAE and RMSE retain arrival scale; MAPE can become unstable for small markets with near-zero values (Hyndman & Koehler, 2006).

The applied gap is that Malaysian research mainly estimates aggregate or long-run relationships, while forecasts rarely integrate monthly source-market arrivals, bilateral exchange rates, global and Malaysian GPR, energy prices, publication lags and post-COVID stress tests. This project fills that gap with a pooled direct model for 20 markets and one- to four-month horizons, tested against transparent benchmarks using only information available at each forecast origin. Direct horizon-specific models also avoid recursively feeding earlier prediction errors into later horizons. External indicators are not presumed to help every horizon; their value is evaluated empirically.

3.0 METHODOLOGY AND DATA ANALYSIS METHODS

3.1 Data Collection and Sources

The study uses a monthly panel of 20 international source markets from March 2017 to September 2026. Monthly arrivals by nationality were obtained from Tourism Malaysia publications produced with the Immigration Department and mapped to stable ISO3 codes (Tourism Malaysia, 2026). Published arrivals end in May 2026; the 80 country-months from June to September 2026 remain blank as the live forecast horizon. Results therefore describe the 20 modelled markets, not total arrivals from every country.

Explanatory data comprise the global and Malaysia-specific Geopolitical Risk indices (Caldara & Iacoviello, 2022), MYR-per-source-currency rates from Bank Negara Malaysia, a documented BNM-Bangladesh Bank cross-rate for BDT (Bangladesh Bank, n.d.; Bank Negara Malaysia, 2026), World Bank Brent prices (World Bank, 2026), and Ministry of Finance fuel prices (Ministry of Finance Malaysia, 2026). DOSM indicators were collected as candidates but excluded from the locked model after publication-lag assessment (Department of Statistics Malaysia, 2026a). The SQLite database is the authoritative pipeline input, while a 63-entry catalogue records provenance, units, transformations and missingness.

3.2 Data Cleaning and Preprocessing

Cleaning controls

Dates, numeric fields and identifiers were standardised, and each market was constrained to one ISO3 code, currency and ASEAN segment. The main correction concerned statistical grain: 69,040 country-day storage rows contained monthly arrivals and exchange rates repeated across days. Code assertions found no conflicting values before one row per country-year-month was retained, producing 2,300 country-months (20 x 115); repeated monthly values were never summed. The 80 unpublished arrival targets were not imputed. Exchange-rate gaps were source-completed from official observations or the documented BDT cross-rate, while original and completed tables were kept separately. Final checks found no duplicate keys, invalid months, negative arrivals, non-positive exchange rates or inconsistent target flags.

3.3 Data Integration and Harmonisation

Monthly integration and forecast alignment

Arrival, exchange-rate, GPR and Brent series retained their published monthly values; weekly fuel levels were converted to calendar-day-weighted monthly means. Country-month records were then joined to one common month-level predictor record using validated many-to-one

keys and consistent units. Publication-lag logic makes the June-September 2026 forecasts use May-August predictor observations. The audit confirmed that all 19 numeric model features were complete for all 80 live rows and that the GPR and Brent values actually used had imputation flags of zero. Provisional September predictors and completed DOSM indices do not enter the locked model. This produces a leakage-controlled panel for the modelling steps in Sections 3.4-3.6.

3.4 Forecasting Problem Formulation

The integrated dataset described in Sections 3.1 to 3.3 provides monthly tourist-arrival counts for 20 major source markets, together with national-level macroeconomic, geopolitical-risk (GPR; Caldara and Iacoviello, 2022), energy-price and exchange-rate indicators. This section defines the forecasting problem that the machine-learning component is designed to address.

Unit of analysis

The raw panel stores one row per source country per calendar day (69,040 rows), but the monthly measures (tourist arrivals and the MYR exchange rate) repeat identically across every day within a month. The true information content is therefore at the country-month level: $20 \text{ countries} \times 115 \text{ months} = 2,300 \text{ observations}$. An integrity assertion in the code verifies this assumption programmatically before any analysis. Collapsing to the country-month grain is a prerequisite. Training or evaluating on the raw daily rows would allow different days of the same month to appear on both sides of a train-test split. This would inflate validation scores through data leakage.

Forecasting task

The model predicts monthly tourist arrivals for each of the 20 source markets at horizons of 1, 2, 3 and 4 months ahead. A single pooled model is used, with the forecast horizon included as an explicit input feature. This direct multi-horizon design was chosen over recursive forecasting, which reuses each prediction as an input for the next step. This avoids propagation of prediction errors across horizons. The design was also preferred over per-country models because the roughly 111 usable monthly observations per market would leave individual models data-starved. A pooled model borrows statistical strength across all 20 series, which is the standard approach for short-panel demand forecasting.

Hold-out structure

Tourist arrivals for June to September 2026 remain unpublished as of the date of this analysis. These 80 country-months ($20 \text{ markets} \times 4 \text{ months}$) constitute the genuine forecast deliverable

and are never used in model training or evaluation. All reported accuracy metrics are based exclusively on historical data with known outcomes.

Coverage scope

All arrival figures in this report refer to the 20 modelled source markets only, which are a subset of Tourism Malaysia’s Top-45 arrivals report (Tourism Malaysia, 2026). They should not be interpreted as Malaysia’s total tourist arrivals from all countries.

3.5 Forecasting Model and Feature Design

The model uses 19 numeric features and one categorical identifier (source country), organised into four groups:

Feature group	Features	Count
Arrivals history	Origin-month actual, lags (1, 2, 3, 6, 12 months), 3-month and 6-month rolling means, year-on-year growth, same-month-last-year reference	10
Exogenous indicators	GPR Global, GPR Malaysia, Brent crude oil, RON97 fuel price, FX rate (MYR per source currency), peninsular diesel price	6
Calendar and structure	Target-month sine and cosine (seasonality encoding), forecast horizon	3
Market identity	Source country (categorical, 20 levels)	1

Publication-lag-aware feature alignment

A central design requirement is that each predictor is sourced only from data that would genuinely have been available at the time of the forecast. Different variables have different publication delays. Exchange rates and fuel prices are effectively real-time (Bank Negara Malaysia daily rates, Bank Negara Malaysia, 2026; Ministry of Finance weekly retail prices, Ministry of Finance Malaysia, 2026). GPR indices and Brent crude are available with approximately one month’s delay. DOSM’s Malaysian Economic Indicators (MEI) series follow a release cycle of roughly two months, verified against DOSM’s own release archive (Department of Statistics Malaysia, 2026a). In that archive, reference month M is published around the 20th to 25th of month M+2. Features were aligned using documented publication schedules where available and conservative lag assumptions where exact release timing could not be verified. The tourist-arrival publication lag determines the ‘simulated present’ at each forecast origin. It is treated as an explicit assumption (primary: 4 months; sensitivity: 3 months), because no public release-date archive was found to verify it.

Feature curation

The initial feature set included all five MEI sub-indices (leading, coincident, lagging, and two diffusion indices). After applying the availability constraints, these indices become increasingly stale at horizons 3 to 4, and their measured importance to the model fell to near zero. They were therefore excluded from the final feature set. This is a deliberate modelling simplification: Malaysia's official leading indicators do not contribute to the final forecast, because their publication timing makes them uninformative at the horizons where the forecast is most needed. This is disclosed as a limitation.

Candidate models

Four models were compared.

- (1) A naive-last baseline repeats the most recent actual arrivals for all horizons.
- (2) A seasonal-naive baseline uses arrivals from the same calendar month one year earlier.
- (3) Ridge regression uses L2 regularisation and explicit $\text{GPR} \times \text{country}$ interaction terms, primarily to provide interpretable per-country GPR coefficients.
- (4) LightGBM (Ke et al., 2017) is a gradient-boosted decision-tree method that handles mixed numeric and categorical inputs natively and captures non-linear relationships without requiring hand-specified interactions.

3.6 Validation Design and Reliability Testing**Rolling-origin backtesting**

Model accuracy is assessed using a rolling-origin backtest with five primary validation origins: March 2023, September 2023, March 2024, September 2024 and September 2025. These origins were chosen to span the post-recovery period with stable tourism conditions. They are spaced at least six months apart so that consecutive validation windows do not heavily overlap. At each simulated origin, the model is trained on all earlier data and used to forecast the following 1 to 4 months. The forecasts are then compared against the actual, published arrivals.

No-leakage rule

Because the pooled horizon table contains rows from different forecast origins whose target months can overlap, a naive origin-based split is not sufficient to prevent data leakage. The validation enforces a row-level rule: a training row is included only if its target month's actual arrivals were already published by the simulated forecast date. This is more precise than an arbitrary fixed gap between training and validation. It excludes less data while still preventing the use of future information and reflects the genuine information constraints a forecaster would face.

COVID stress testing

A separate set of five validation origins (March 2020 through March 2022) is evaluated under the same protocol. These folds deliberately span the COVID-19 border closure and the subsequent reopening. During this period, total arrivals across the 20 markets fell from approximately 2 million per month to near zero and remained suppressed for nearly two years. The stress-test results are reported separately from the primary evaluation and are not averaged into it. This allows model performance under normal conditions and under an extreme structural disruption to be assessed separately.

Metrics

Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are reported per horizon. Percentage-based metrics such as MAPE were avoided because several smaller source markets (for example, Nepal, Myanmar and Pakistan) have monthly arrival volumes near zero during parts of the sample. This inflates percentage errors and makes them unreliable as a basis for model comparison.

Training-eligibility parity. Backtesting and deployment apply the same eligibility condition: a row is used only if it has a full 12-month arrivals history. This ensures that the procedure evaluated historically is the same procedure used to generate the published forecasts.

Sensitivity analysis

Because the tourist-arrival publication lag is an assumption rather than a verified fact, the full backtest is repeated under an alternative lag of 3 months. The purpose is to confirm that the model-selection conclusion does not depend on the specific lag value chosen.

4.0 FINDINGS AND RESULTS

4.1 Forecast Accuracy and Model Selection

Table 1 reports the primary backtest results under the 4-month arrival-lag assumption. All figures are mean absolute error in arrivals (persons), averaged across the five primary validation folds.

Table 1. *Primary backtest MAE by model and forecast horizon*

Horizon	Naive-last	Seasonal-naive	Ridge	LightGBM
1 (1 month ahead)	12,752	23,802	12,117	11,817
2 (2 months ahead)	9,671	20,404	9,757	9,571
3 (3 months ahead)	25,157	21,191	24,258	21,504
4 (4 months ahead)	21,171	25,413	20,775	20,983

No single model is best at every horizon. LightGBM achieves the lowest MAE at horizons 1 and 2, with values of 11,817 and 9,571. The seasonal-naive baseline is marginally best at horizon 3 (21,191 against LightGBM's 21,504). Ridge is marginally best at horizon 4 (20,775 against 20,983). The margins at horizons 2 to 4 range from 100 to 313 arrivals, which is under 2% of the corresponding error levels. With only five validation folds, the evidence is not sufficient to establish a reliable ordering between the models at these horizons. These per-horizon differences should therefore not be read as meaningful separations. The clearer separation is against the seasonal-naive baseline at horizons 1, 2 and 4, where LightGBM reduces error by 50%, 53% and 17% respectively.

Model selection

Because the deployed design uses one pooled model across all four horizons, the selection criterion is aggregate error across all primary backtest predictions rather than per-horizon wins. On that criterion, LightGBM has the lowest aggregate MAE (15,969), ahead of Ridge (16,727), naive-last (17,188) and seasonal-naive (22,702). It also has the lowest aggregate RMSE (36,043). This criterion matches the deployed design because a single model generates all forecasts, so total error across all predictions is the quantity that matters operationally. The choice is defensible rather than decisive. On rank-based criteria, Ridge is ahead. It wins 7 of the 20 fold-by-horizon cells against LightGBM's 4 and achieves a slightly better mean rank (2.15 against 2.30). LightGBM also has the larger worst-case cell error. Ridge is therefore a legitimate alternative, and LightGBM is reported here as the model that minimises total forecast error under the pooled design, not as clearly superior. A champion-by-horizon strategy was rejected because the per-horizon margins are small enough that switching models per horizon would risk fitting fold-level noise.

Table 2. *COVID/reopening stress-test MAE (selected models, 4-month lag assumption)*

Horizon	Naive-last	LightGBM
1	9,647	12,805
2	12,366	16,279
3	15,511	20,730
4	16,134	20,334

Under the COVID structural break, naive persistence (simply repeating the most recent month's arrivals) achieves the lowest MAE at all four horizons. This is not a failure of the machine-learning approach. It reflects a fundamental limitation: none of the models tested anticipated a discrete, unprecedented policy shock such as a full border closure. The stress-test result is reported transparently because it supports a practical insight. Any operational forecasting system should present horizon-specific reliability indicators rather than a single accuracy figure, so that decision-makers can calibrate their confidence appropriately.

Sensitivity to the arrival-lag assumption

Repeating the full backtest under a 3-month lag assumption produces LightGBM MAE values of 12,611, 10,762, 20,835 and 20,509 for horizons 1 to 4 respectively. Under the primary 4-month assumption, the corresponding values are 11,817, 9,571, 21,504 and 20,983. The differences range from 2% to 12%, and at horizons 3 to 4 the 3-month assumption is in fact marginally better. Crucially, the model-selection conclusion is unchanged: LightGBM retains the lowest aggregate error under both assumptions.

Feature importance

The five most influential inputs to the final LightGBM model were identified using split-count importance. The top three are (1) most recent actual arrivals at the forecast origin, (2) same-month-last-year arrivals and (3) Brent crude oil price (World Bank Commodity Markets, 2026). Source country identity ranks fourth, followed by RON97 fuel price. The GPR Global Index ranks sixth. This ordering is consistent with the expectation that recent arrival history is the strongest short-term predictor, while external economic and geopolitical conditions contribute additional information at longer horizons where historical persistence weakens.

4.2 Solution Approach

June to September 2026 forecast

The final LightGBM model, trained on all available history (7,720 origin-horizon training samples, drawn from 1,960 unique country-month origins), produces the following aggregate forecasts for the 20 modelled source markets:

Table 3. *Forecast total arrivals across 20 modelled source markets, June to September 2026*

Target month	Horizon	Forecast arrivals (20 markets)
June 2026	1	2,278,401
July 2026	2	2,287,744
August 2026	3	2,173,788
September 2026	4	2,054,893

The forecast indicates a broadly stable arrival volume in June to July, followed by a gradual decline into August to September. This pattern is consistent with the seasonal profile observed in prior years. The five largest forecast source markets for June 2026 are Singapore (994,700), China (358,309), Indonesia (312,865), Thailand (134,150) and India (108,880). These figures represent the model’s point forecast only. No prediction intervals are produced, so the forecast should be treated as an estimated trend rather than a precise bound.

Exogenous data completeness

The exogenous predictors used by the final model (GPR, Brent crude, fuel prices and exchange rates) are fully observed for all source months used across horizons 1 to 4 (May to August 2026). The MEI indices, which would have been partly imputed at longer horizons, were excluded from the final feature set (Section 3.5), so this concern does not apply to the deployed model. Verification of the horizon-4 feature row confirms that its fuel and exchange-rate inputs are sourced from August 2026, a complete month with full price coverage. The incomplete September 2026 month is not used as a feature source at any horizon. No residual data-completeness caveat therefore applies to the exogenous inputs of the published forecast.

GPR and Source-Market Heterogeneity

A central question of this study is whether the association between geopolitical risk and tourist arrivals varies across source markets. Because the GPR index is a national-level indicator that is identical for all 20 countries within a given month, it cannot, on its own, explain cross-country differences. To identify market-specific associations, we employ two complementary approaches: a Ridge regression with explicit $\text{GPR} \times \text{country}$ interaction terms, and SHAP-based decomposition of the LightGBM model.

Ridge $\text{GPR} \times \text{country}$ interaction

A Ridge regression was fitted on the same feature set, with $\text{GPR} \times \text{country}$ interaction terms and the United States as the reference category. The United States is omitted from both the country dummies and the interactions so that all coefficients are uniquely identified. Coefficients are recovered in raw feature space before the base and interaction terms are combined because the standardised base and interaction columns have different scales and cannot validly be added. The recovered slopes were then verified independently by perturbation. GPR was raised by exactly one standard deviation while all other inputs were

held fixed, and this reproduced every reported slope exactly across all 20 markets. The results show that 16 of 20 source markets have a negative controlled association between GPR and arrivals. Higher GPR is associated with lower predicted arrivals after controlling for arrivals history, seasonality, oil prices, exchange rates and all other features. Singapore, Indonesia and China have positive slopes, and Thailand is approximately neutral.

Table 4. *Controlled GPR slope by source market, in arrivals per one standard-deviation increase in GPR (Ridge, USA as reference category)*

Market	Segment	GPR slope
Singapore	ASEAN	+38,626
Indonesia	ASEAN	+8,175
China	Non-ASEAN	+6,751
Thailand	ASEAN	+297
India	Non-ASEAN	-563
USA (reference)	Non-ASEAN	-2,084
Philippines	ASEAN	-3,168
Japan	Non-ASEAN	-4,102
Germany	Non-ASEAN	-4,666
South Korea	Non-ASEAN	-4,669
Nepal	Non-ASEAN	-4,807

Note: Table shows the 5 highest and 6 lowest slopes; the full 20-market table is available in the supplementary outputs. Slopes are expressed in arrivals per one standard-deviation increase in the GPR Global Index and reflect partial associations after controlling for all other features in the model.

Interpretation and caveats

Two important caveats apply. First, the raw (unconditional) correlation between GPR and arrivals across all country-months is weakly positive (+0.08), while the controlled Ridge slope for most markets is negative. What the evidence establishes is that the estimated association changes sign once the other predictors are included; it does not identify the mechanism behind that change. A plausible contributor is the shared time trend. GPR trends upward over the sample period (correlation with time: +0.54), while arrivals recovered post-COVID. An unadjusted comparison may therefore partly reflect co-movement in time rather than a relationship between the two series. Correlation among the predictors themselves may also contribute. These remain candidate explanations rather than a demonstrated mechanism. Second, these results are associational, not causal. The regression identifies statistical co-movement between GPR and arrivals within the model, but does not establish that changes in geopolitical risk cause changes in tourist arrivals. Causal inference would require an

identification strategy (such as an instrumental variable or a natural experiment) that is beyond the scope of this study.

SHAP model-contribution analysis

As a complementary perspective, SHAP (SHapley Additive exPlanations; Lundberg and Lee, 2017) values were extracted from the final LightGBM model to quantify how much each feature contributes to individual predictions. The mean absolute SHAP value for the GPR Global Index was grouped by source country and normalised by each market's average arrival volume. The ranking shifts substantially compared with the raw (un-normalised) values. Smaller markets such as Nepal, Myanmar and Pakistan show the highest proportional GPR contribution. Large markets such as Singapore and China rank highest in absolute terms because any factor's contribution scales with the magnitude of the prediction. The normalised ratios should be interpreted with caution for small-volume markets, where the denominator is small and the ratio correspondingly noisy. Two limits apply to how these values should be read. They are computed on the observations used to fit the final model. Therefore, they describe the behaviour of the fitted model on its training data rather than validated behaviour on unseen periods. An out-of-sample explainability analysis was not undertaken at this stage. They also measure the model's internal attribution, not real-world sensitivity, and are reported here as evidence of model behaviour rather than as causal findings.

5.0 DASHBOARD OUTPUT

5.1 Dashboard Overview

The dashboard was developed using Microsoft Power BI to present an interactive analysis of Malaysia's tourism performance and tourist arrival forecasts. It consists of four main pages: Malaysia Tourism Overview, Market Analysis, Tourism Drivers, and Forecast, each offering a distinct perspective on the tourism industry, ranging from historical trends and market segmentation to external tourism drivers and future arrival predictions.

The Malaysia Tourism Overview page summarises Malaysia's recent tourism performance, covering 20 tracked source markets. Key performance indicators are presented at the top of the dashboard, including the latest monthly tourist arrivals, year-on-year (YoY) growth, year-to-date (YTD) tourist arrivals, and YTD growth. The dashboard uses May 2026 as the latest actual month, with YTD figures covering January to May 2026. As of this period, YTD arrivals reached 9.95 million, representing a 0.9% YoY increase, while May 2026 alone recorded 1.96 million arrivals, a 3.11% YoY decline, indicating softer month-on-month momentum despite stable YTD growth. Historical tourist arrival trends are further presented through monthly and annual visualisations.

The Market Analysis page focuses on the source markets of tourists visiting Malaysia, identifying the top source countries and comparing arrivals from ASEAN and non-ASEAN markets. This allows users to understand the major markets contributing to Malaysia's tourism industry and the relative importance of regional versus international visitors.

The Tourism Drivers page examines the relationship between tourist arrivals and several external factors, including Brent crude oil prices, exchange rates, and the Global Geopolitical Risk (GPR) Index. By comparing these factors with tourist arrivals, the dashboard provides a broader view of how economic and geopolitical conditions may influence tourism demand.

The Forecast page predicts Malaysia's tourist arrivals for June to September 2026, based on 20 modelled source markets and following the latest actual data available up to May 2026. The page also evaluates forecasting performance through backtesting, feature importance, model comparison, COVID stress testing, and lag sensitivity analysis.

5.2 Dashboard Functions and Interactivity

The dashboard was designed to provide users with an interactive way to explore Malaysia's tourism data. Power BI's filtering and interactive features allow users to examine the data from different perspectives without manually altering the underlying dataset.

On the Malaysia Tourism Overview page, users can apply the Market Segment (ASEAN / Non-ASEAN) and Source Country slicers to filter the dashboard according to their preferred tourism market or country of interest. For example, users can select a specific source country to examine its individual arrival trend and compare it against overall market performance.

The dashboard also supports interactive cross-filtering across visualisations. When a user selects a category or data point in one visual, related visuals respond accordingly, making it easier to identify relationships and patterns between different tourism indicators.

The Tourism Drivers page similarly includes a Source Country filter, allowing users to examine tourism performance alongside external factors such as Brent crude oil prices, exchange rates, and geopolitical risk indicators, either at the aggregate level or for a specific market.

The Forecast page provides interactive model evaluation. Users can compare different forecasting models based on their Mean Absolute Error (MAE) and examine how model performance changes across different forecast horizons. The dashboard also includes a COVID stress test and a LightGBM lag sensitivity analysis, allowing users to evaluate model robustness under different conditions.

These interactive functions improve the usability of the dashboard, as users can move from a general overview to more detailed, specific analysis based on their interests. The dashboard therefore functions not only as a static presentation of results but also as an analytical tool for exploring tourism trends and forecasting performance.

5.3 Dashboard Visuals

The dashboard contains several types of visualisations, including cards, line charts, bar charts, combination charts, donut charts, slicers, and model evaluation charts. Different visual types are used to communicate different aspects of the tourism analysis.

Malaysia Tourism Overview

The first page uses KPI cards to display the latest tourist arrival figures and growth rates. As of the latest update, YTD arrivals (January–May 2026) reached 9.95 million, representing a 0.9% YoY increase, while May 2026 alone recorded 1.96 million arrivals, a 3.11% YoY decline, indicating softer month-on-month momentum despite stable YTD performance. These cards provide users with a quick understanding of recent tourism performance across the 20 modelled markets. A line chart presents the monthly tourist arrival trend through May 2026, allowing users to observe recurring seasonal peaks and troughs, such as the year-end surges visible around December each year.

A bar chart presents the Top 5 Source Countries for 2026 YTD, led by Singapore (3.47M), followed by China (1.87M), Indonesia (1.47M), Thailand (0.71M), and India (0.48M), which allows the major tourism markets to be identified easily. In addition, a combination chart presents annual tourist arrivals and YoY growth for complete years (2018–2024), providing a longer-term view of tourism performance, including the sharp decline during the COVID-19 period and the subsequent recovery. A dedicated scope note at the top of the page reminds users that all figures reflect the 20 modelled source markets rather than Malaysia's total national arrivals, avoiding potential misinterpretation of the headline numbers.

Market Analysis

The Market Analysis page uses a horizontal bar chart to present the Top 10 Source Countries, making it easier to compare the relative contribution of different source markets. A donut chart compares ASEAN and non-ASEAN tourist arrivals, providing a simple visual representation of Malaysia's continued reliance on regional tourism demand.

Tourism Drivers

The Tourism Drivers page uses combination charts to compare tourist arrivals with Brent crude oil prices, exchange rates, and the Global Geopolitical Risk Index. These charts allow changes in tourist arrivals to be visually compared with movements in external economic and geopolitical factors over time, such as the notable decline seen around 2020.

A separate line chart presents the Average Malaysia GPR Index on an annual basis, showing how domestic geopolitical risk has evolved over the observed years. A Source Country slicer on this page allows users to isolate a specific market for closer examination. Both driver visuals carry a brief note clarifying that the patterns shown reflect descriptive co-movement only and should not be interpreted as causal relationships.

Forecast

The Forecast page presents Malaysia's predicted tourist arrivals for June to September 2026 across the modelled source markets. The Forecast Tourist Arrivals chart shows how projected arrivals move over the forecast period, indicating a gradual softening in demand toward the end of it.

The Backtest chart compares actual arrivals with the model's predictions across past months, allowing prediction accuracy to be assessed visually, alongside a summary KPI reporting the overall backtest error.

The Feature Importance chart shows which variables most influence the model's predictions, with recent arrival history and external factors such as oil prices among the top contributors.

The dashboard also compares several forecasting models using MAE across different forecast horizons, where a lower MAE indicates better performance.

Finally, the COVID Stress Test chart evaluates how each model performs under an extreme shock scenario, showing that naive persistence can be more robust than rigid seasonal baselines during structural breaks. The Lag Sensitivity chart compares different lag configurations across forecast horizons. Together, these visuals provide additional evidence of the reliability and robustness of the forecasting models under different conditions.

6.0 CONCLUSION

This study evaluated the relationship between geopolitical risk and international tourism demand in Malaysia and developed a short-term forecasting framework for the 20 modelled source markets. A monthly country panel was built covering arrivals, exchange rates, geopolitical risk, and energy prices, and a pooled LightGBM model was developed and benchmarked against naive, seasonal-naive, and Ridge-regression baselines, with results communicated through an interactive Power BI dashboard.

Among the models tested, LightGBM achieved the lowest aggregate forecast error, making it the most suitable choice for generating forecasts across all horizons, although Ridge regression remained a reasonable alternative. A stress test using the COVID period showed that none of the models could anticipate a sudden, unprecedented shock such as a full border closure, which is an important limitation to keep in mind when interpreting the forecasts. Additional sensitivity testing showed that the overall model choice remained stable under different assumptions, supporting the robustness of the results.

On geopolitical risk, the analysis found that its association with tourist arrivals varies by market: most source markets show a negative association, while a few, including Singapore, Indonesia, and China, show a positive one. This suggests that geopolitical risk does not affect all markets in the same way, and that findings based on national totals alone could miss these differences. It is important to note that these results reflect statistical associations rather than causal relationships.

The resulting forecast for June to September 2026 suggests relatively stable arrivals in June and July, followed by a gradual decline toward September, consistent with typical seasonal patterns. Overall, the study shows that an interpretable machine-learning approach can support short-term tourism forecasting for Malaysia while remaining transparent about its limitations. Future work could explore prediction intervals, additional shock scenarios, and stronger methods for establishing causal relationships between geopolitical risk and tourism demand.

REFERENCES

- Bangladesh Bank. (n.d.). Exchange rate of Taka. Bangladesh Bank.
<https://www.bb.org.bd/en/index.php/econdata/exchangerate>
- Bank Negara Malaysia. (2026). Exchange rates. Bank Negara Malaysia.
<https://www.bnm.gov.my/exchange-rates>
- Becken, S., & Lennox, J. (2012). Implications of a long-term increase in oil prices for tourism. *Tourism Management*, 33(1), 133-142.
<https://doi.org/10.1016/j.tourman.2011.02.012>
- Boto-Garcia, D. (2025). The elasticity of tourist arrivals to oil prices. *Tourism Economics*. Advance online publication. <https://doi.org/10.1177/13548166251364305>
- Caldara, D., & Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4), 1194-1225. <https://doi.org/10.1257/aer.20191823>
- Department of Statistics Malaysia. (2026a). Malaysian Economic Indicators — Leading, Coincident and Lagging Index. Department of Statistics Malaysia.
<https://www.dosm.gov.my>
- Department of Statistics Malaysia. (2026b, September 15). Tourism satellite account 2025.
<https://www.dosm.gov.my/portal-main/release-content/tourism-satellite-account-2025>
- Hampton, M. P., Jeyacheya, J., & Nair, V. (2023). Post-COVID tourism revealed: Evidence from Malaysia. *Annals of Tourism Research*, 103, 103671.
<https://doi.org/10.1016/j.annals.2023.103671>
- Hyndman, R. J., & Koehler, A. B. (2006). Another look at measures of forecast accuracy. *International Journal of Forecasting*, 22(4), 679-688.
<https://doi.org/10.1016/j.ijforecast.2006.03.001>
- Imamboccus, R., Seetanah, B., Nunkoo, R., & Jaffur, Z. K. (2024). The impact of exchange rate and exchange rate volatility on tourism demand using disaggregated data. *International Journal of Tourism Research*, 26(2), e2640.
<https://doi.org/10.1002/jtr.2640>
- Karimi, M. S., Khan, A. A., & Karamelikli, H. (2019). Asymmetric effects of real exchange rate on inbound tourist arrivals in Malaysia: An analysis of price rigidity. *International Journal of Tourism Research*, 21(2), 156-164.
<https://doi.org/10.1002/jtr.2249>

- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems* 30 (pp. 3146-3154). Curran Associates.
- Lee, C.-C., Olasehinde-Williams, G., & Akadiri, S. S. (2021). Geopolitical risk and tourism: Evidence from dynamic heterogeneous panel models. *International Journal of Tourism Research*, 23(1), 26-38. <https://doi.org/10.1002/jtr.2389>
- Long, W., Liu, C., & Song, H. (2019). Pooling in tourism demand forecasting. *Journal of Travel Research*, 58(7), 1161-1174. <https://doi.org/10.1177/0047287518800390>
- Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30 (pp. 4765-4774). Curran Associates.
- Makridakis, S., Spiliotis, E., & Assimakopoulos, V. (2022). M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4), 1346-1364. <https://doi.org/10.1016/j.ijforecast.2021.11.013>
- Ministry of Finance Malaysia. (2026). Retail fuel prices. Ministry of Finance Malaysia. <https://www.mof.gov.my>
- Ministry of Tourism, Arts and Culture Malaysia. (2020). National Tourism Policy 2020-2030. <https://www.motac.gov.my/en/category/download/dasar-pelancongan-negara-dpn-2020-2030/>
- Montero-Manso, P., & Hyndman, R. J. (2021). Principles and algorithms for forecasting groups of time series: Locality and globality. *International Journal of Forecasting*, 37(4), 1632-1653. <https://doi.org/10.1016/j.ijforecast.2021.03.004>
- Papagianni, E., Evgenidis, A., Tsagkanos, A., & Megalooikonomou, V. (2024). Tourism demand in the face of geopolitical risk: Insights from a cross-country analysis. *Journal of Travel Research*, 63(8), 2094-2119. <https://doi.org/10.1177/00472875231206539>
- Peng, B., Song, H., & Crouch, G. I. (2014). A meta-analysis of international tourism demand forecasting and implications for practice. *Tourism Management*, 45, 181-193. <https://doi.org/10.1016/j.tourman.2014.04.005>
- Seetaram, N. (2010). Computing airfare elasticities or opening Pandora's box. *Research in Transportation Economics*, 26(1), 27-36. <https://doi.org/10.1016/j.retrec.2009.10.005>
- Song, H., & Li, G. (2008). Tourism demand modelling and forecasting: A review of recent research. *Tourism Management*, 29(2), 203-220. <https://doi.org/10.1016/j.tourman.2007.07.016>

- Tan, C.-Y., & Soon, S.-V. (2023). Tourism demand for Malaysia: Further evidence from panel approaches. *Asia Pacific Management Review*, 28(4), 459-469.
<https://doi.org/10.1016/j.apmrv.2022.12.006>
- Tashman, L. J. (2000). Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4), 437-450. [https://doi.org/10.1016/S0169-2070\(00\)00065-0](https://doi.org/10.1016/S0169-2070(00)00065-0)
- Tourism Malaysia. (2025). Visitor performance to Malaysia January to December 2024 [Infographic].
https://data.tourism.gov.my/frontend/pdf/Infographic_visitor_performance_to_malaysia_jan-dec_2024.pdf
- Tourism Malaysia. (2026). Monthly tourist arrivals report. Tourism Malaysia, in cooperation with the Immigration Department of Malaysia.
- United Nations Department of Economic and Social Affairs. (n.d.). Sustainable tourism. United Nations Sustainable Development. <https://sdgs.un.org/topics/sustainable-tourism>
- World Bank. (2026). Commodity Markets — Pink Sheet. World Bank Group.
<https://www.worldbank.org/en/research/commodity-markets>